

Audio Convolution and Inverse Filtering for Background Music Suppression

1) Methodology

The following steps were performed:

- Loaded the audio file and converted it to mono.
- Normalized the audio samples.
- Designed five impulse responses manually:
 - Identity Filter
 - Low-Pass Filter
 - High-Pass Filter
 - Band-Pass Filter
 - Band-Stop Filter
- Applied convolution between the audio signal and each impulse response.
- Performed inverse filtering using FFT.
- the original, filtered, and recovered audio.
- Analyzed the effect of changing the cutoff frequencies.

2) Experimental Observations

Identity Filter

- No noticeable change in the audio.
- Original and recovered signals were almost identical.
- No background music suppression was observed.

Low-Pass Filter

- High-frequency components were reduced.
- The audio became smoother and slightly muffled.
- Background music suppression was minimal.

High-Pass Filter

- Reduced bass and low-frequency instruments.
- Vocals became relatively clearer at moderate cutoff frequencies.
- Increasing the cutoff frequency from approximately 1250 Hz toward 3500 Hz reduced vocal clarity because more vocal frequencies were removed.
- Higher cutoff frequencies introduced noticeable noise and distortion.
- In some cases only high-frequency background instruments remained audible.

Band-Pass Filter

- Produced only minor audible changes.
- Vocals and instruments were both preserved because they occupy similar frequency ranges.
- No significant background music suppression was achieved.

Band-Stop Filter

- Produced one of the most noticeable changes.
- Successfully attenuated the selected frequency band.
- Depending on the stop-band selection, vocals or instruments became weaker.
- This filter gave better audible suppression than the Low-Pass and Band-Pass filters.

3) Effect of Changing Cutoff Frequencies

High-Pass Filter

- Lower cutoff frequencies preserved most of the vocal content.
- Increasing the cutoff frequency progressively removed low-frequency instruments.
- Very high cutoff frequencies also removed significant portions of the vocal signal, making the voice weak while leaving only high-frequency instrumental sounds.
- Higher cutoff frequencies increased the perceived noise after inverse filtering.

Band-Stop Filter

- Different stop-band ranges produced different suppression effects.
- Wider stop bands removed larger portions of the audio spectrum.
- Narrow stop bands produced subtler changes.
- Appropriate selection of the stop-band frequencies resulted in the most noticeable alteration of background accompaniment.

4) Analysis

The High-Pass filter was effective in reducing bass components, making vocals relatively more noticeable in certain sections. However, increasing the cutoff frequency excessively also attenuated important vocal frequencies and introduced additional noise.

The Band-Stop filter produced the most significant audible changes because it selectively attenuated a chosen frequency range. Nevertheless, since vocals and musical instruments overlap in frequency, complete isolation of either component was not possible.

The Low-Pass and Band-Pass filters produced comparatively minor changes. Both vocals and instruments occupy similar frequency ranges, so these filters affected both simultaneously instead of isolating one from the other.

Simple kernels such as the Identity, Echo, Alternating, and Difference filters demonstrated the principle of convolution but were not suitable for background-music suppression.

Inverse filtering successfully reconstructed the original signal for simple filters. However, reconstruction quality deteriorated for more selective filters because frequencies with very small magnitudes amplified numerical errors and noise.

5) Limitations

- Vocals and background instruments overlap in frequency.
- Linear FIR filters cannot distinguish between different sound sources.
- Inverse filtering amplifies noise when the filter response approaches zero.

6) Conclusion

This experiment demonstrated the application of convolution and inverse filtering using several manually designed impulse responses. Among all the filters tested, the High-Pass and Band-Stop filters produced the most noticeable changes in the audio signal. The High-Pass filter effectively reduced low-frequency musical components, although increasing the cutoff frequency beyond an optimal value weakened the vocals and introduced noise. The Band-Stop filter altered the balance between vocals and background music more significantly than the other filters, depending on the selected stop-band frequencies.

Overall, the experiment demonstrates that convolution and inverse filtering are valuable tools for frequency-selective signal processing but are insufficient for complete background-music removal because vocals and instruments share overlapping frequency ranges. More advanced techniques, such as adaptive filtering, spectral subtraction, or machine learning-based source separation, are required for effective vocal and accompaniment separation.